

Studying the Effect of Pre-Polymer Composition and Incorporation of Surface-Modifying Amphiphilic Additives on the Fouling-Release Performance of Amphiphilic Siloxane-Polyurethane Coatings

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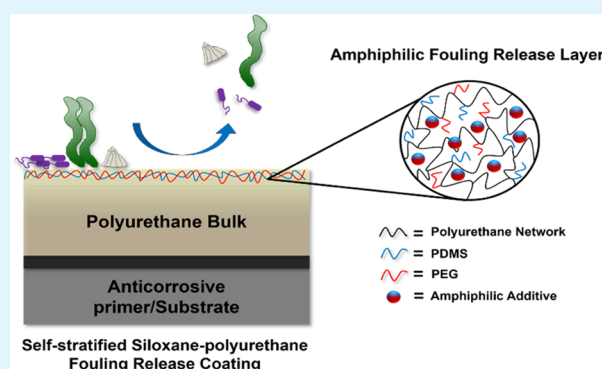
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ABSTRACT: Combining amphiphilic fouling-release (FR) coatings with the surface-active nature of amphiphilic additives can improve the antifouling/fouling-release (AF/FR) properties needed to offer broad-spectrum resistance to marine biofouling. This work is focused on further tuning the amphiphilic character of a previously developed amphiphilic siloxane-polyurethane (SiPU) coating by varying the amount of PDMS and PEG in the base system. Furthermore, surface-modifying amphiphilic additives (SMAAs) were incorporated into these amphiphilic FR SiPU coatings in varying amounts. ATR-FTIR, contact angle and surface energy measurements, and AFM were performed to assess changes in surface composition, wettability, and morphology. AF/FR properties were evaluated using laboratory biological assays involving *Cellulophaga lytica*, *Navicula incerta*, *Ulva linza*, *Amphibalanus amphitrite*, and *Geukensia demissa*. The surfaces of these coatings varied significantly upon changes in PDMS and PEG content in the coating matrix, as well as with changes in SMAA incorporation. AF/FR properties were also significantly changed, with formulations containing the highest amounts of SMAA showing very high removal properties compared to other experimental formulations, in some cases better than that of commercial standard FR coatings.

KEYWORDS: amphiphilic additives, siloxane-polyurethane, fouling-release coatings, amphiphilic coatings, marine coatings



INTRODUCTION

A common problem that faces structures submerged in seawater is that of marine biofouling. This phenomenon involves the settlement and growth of over 4000 different micro- and macrofouling organisms on these structures, which leads to a variety of negative impacts, especially on the hulls of seafaring vessels.^{1,2} Accumulation of these foulants on ships' hulls causes increased drag and fuel consumption, overall higher economic cost of operation, and greater greenhouse gas emissions, while also promoting highly detrimental invasive species transfer.^{3–6} It is estimated that the United States Navy spends nearly \$6 million dollars per year for their mid-sized vessels such as destroyers, with projections of over 1 billion dollars after a period of 15 years to contend with biofouling.³ A major reason for such a high cost of maintenance of these marine vessels is the highly complex and dynamic nature of the phenomenon of biofouling along with the numerous number of foulants each having various sizes and modes of adhesion.⁷ Several different stages are involved in this process, which usually begins with a "conditioning" layer made up largely of proteins, nutrients, and other small molecules. Other stages that follow include bacterial or microalgae biofilms, which often lead to the

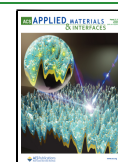
settlement of macrofouling organisms such as macroalgae, barnacles, mussels, and tunicates. However, being dynamic in nature, the amount, and subsequent timing, of each fouling stage varies greatly, making protection against marine biofouling a difficult task for any single approach.^{2,7}

Methods of preventing marine biofouling have undergone numerous changes ever since the navigation of the world's oceans began.² Layers of copper alloy, lead sheaths, and mixtures of tar and hot pitch were widely used on wooden ships as antifouling (AF) countermeasures.⁸ However, as steel ships became more prevalent, along with the advances in materials science at the advent of the 20th century, methods including the use of resins or biodegradable polymer matrices containing biocides became widely accepted.⁹ Arguably the most successful biocides utilized in these coatings were alkyl

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tin-containing paints notably those with tributyltin (TBT). These antifouling (AF) paints were widely used until a worldwide ban was ratified by the International Maritime Organization (IMO) in 2003 due to concerns of their high toxicity toward nontargeted organisms and overall impact on local aquatic ecosystems.¹⁰ Currently, antifouling paint formulations mostly contain biocidal pigments based on copper and zinc oxides which, while effective, are still seen to accumulate in local aquatic environments, leading to increased scrutiny regarding toxicity concerns.¹¹ Due to the constant concerns for use of biocides as a fouling deterrent, nontoxic solutions involving fouling-release (FR) coatings have been widely studied. These FR coatings function by preventing the strong adhesion of marine foulants to surfaces so that under hydrodynamic forces, the organisms will be easier to remove.^{2,7} Several traditional FR systems have been utilized that are composed of low-surface-energy materials such as polydimethylsiloxane (PDMS) or polyfluoroalkyl (PFA) elastomers. However, these coatings often suffer from poor durability, poor mechanical properties, and difficult adhesion to substrates, often needing tie-coats to get the desired protection.^{1,2,7} To remedy these issues, self-stratifying siloxane-polyurethane (SiPU) FR coatings were developed and showed a significant improvement in mechanical properties, adhesion, and durability, while maintaining excellent FR properties.^{12,13} However, the surfaces of these coatings were hydrophobic to which some organisms prefer to settle, limiting their performance compared to more fouling-resistant surfaces.^{14–16}

To address some of the issues with using hydrophobic FR coating systems, amphiphilic FR coatings have been investigated, which incorporate a hydrophilic component to offer fouling resistance, while also maintaining the good release properties of hydrophobic surfaces.^{17–19} There are typically two methods to achieve such a heterogeneous, amphiphilic surface. One method is to alter the composition of a coating system using reactive hydrophilic and hydrophobic moieties. The most widely used hydrophilic component is poly(ethylene glycol) (PEG), while the hydrophobic component is often PDMS or PFA polymers.^{17,20–22} Galhenage et al. explored coatings that involved synthesizing amphiphilic isocyanate pre-polymers containing PEG and PDMS chains of varying molecular weight and amount, and then reacting them further to form an amphiphilic SiPU network.²³ These coatings provided broad-spectrum FR performance for several different micro- and macrofoulants that were comparable to commercial standard coatings such as Intersleek 1100SR and Hempasil X3. Other amphiphilic coating systems being investigated include zwitterionic moieties,^{24–29} polyelectrolyte systems,^{30,31} and those containing peptide mimicking “peptoid” molecules.^{32,33}

Another method of imparting amphiphilic character to a coating's surface is by using surface-active additives that have both hydrophilic and hydrophobic components. Again, PEG and PDMS are the most widely used hydrophilic and hydrophobic moieties being used in several different coatings systems.^{34–38} Zwitterionic-based co-polymers,^{39–41} poly(vinylpyrrolidone)-containing additives,⁴² and poly(oxazoline)-based amphiphilic additives have also been used to improve the AF/FR properties of coating systems.⁴³ Recently, Benda et al. prepared a series of surface-modifying amphiphilic additives (SMAAs) that varied in molecular weight of grafted PEG chains, as well as the degree of grafting onto a polysiloxane backbone. These SMAAs were incorporated into a hydrophobic SiPU coating at varying amounts. It was shown

that the molecular weight of grafted PEG chains had the largest effect on FR performance, followed by the degree of grafting of PEG chains.³⁷ However, this SMAA system, along with the vast majority of coatings that incorporate amphiphilic additives, employs hydrophobic coating matrices. Incorporating these amphiphilic additives into inherently amphiphilic coating matrices could further increase AF/FR performance.

In this study, further investigation of previously prepared amphiphilic isocyanate pre-polymers, and their subsequent amphiphilic coatings, was performed. Pre-polymers were prepared using isophorone diisocyanate polyisocyanate reacted with monocarbinol-terminated 10,000 g/mol PDMS, and hydroxy-terminated 750 g/mol PEG. The equivalents of isocyanate/hydroxyl groups were kept at 3:2, but the amount of PDMS or PEG in the pre-polymer was varied. FTIR, isocyanate titrations, and GPC were utilized to determine the remaining isocyanate content, variation in PDMS and PEG content, as well as the general distribution of pre-polymer species. These pre-polymers were then mixed with additional polyisocyanate, an acrylic polyol, and varying amounts of an SMAA to generate amphiphilic SiPU FR coatings. Water contact angle (WCA) and methylene iodide contact angle (MICA) were measured, and surface energy was determined for each experimental formulation before and after 28 days of immersion in circulating tap water. ATR-FTIR and AFM were utilized to determine the presence of PDMS and PEG at the surface, as well as changes in surface morphology when the amount of PDMS and PEG was changed in the pre-polymer and the amount of incorporated SMAA was increased. Finally, the FR properties of these coatings were evaluated using laboratory biological assays involving the marine bacterium *Cellulophaga lytica*, the diatom *Navicula incerta*, the green macroalga *Ulva linza*, the barnacle *Amphibalanus amphitrite*, and the mussel *Geukensia demissa*.

EXPERIMENTAL SECTION

Materials. The solvents toluene, HPLC-grade tetrahydrofuran (THF), methyl amyl ketone (MAK), isopropanol, ethyl-3-ethoxypropionate (EEP), and acetone were purchased from Sigma-Aldrich. PEG-750, a 750 $\bar{M}_n$ hydroxyl-terminated poly(ethylene glycol) monomethyl ether, platinum(0)-1,3-divinyl-1,1,3,3-tetramethyldisiloxane complex solution with ~2% Pt in xylenes (Karstedt's catalyst), dibutyltin diacetate (DBTDac), acetylacetone, dibutylamine, and anhydrous benzyl alcohol were also purchased from Sigma-Aldrich. Monocarbinol-terminated polydimethylsiloxane (PDMS) with a molecular weight of 10,000 g/mol (MCR-C22) was purchased from Gelest, Inc. Monoallyl-terminated poly(ethylene glycol) monomethyl ether with a molecular weight of 350 g/mol (Polyglykol 350 AM) was provided by Clariant. Polyisocyanate Desmodur Z 4470 BA was provided by Covestro LLC. Hydrochloric acid (0.1 N) and potassium bromide optical disks used for Fourier transform infrared spectroscopy (FTIR) experiments were purchased from Alfa Aesar. For use in coatings formulations, an acrylic polyol consisting of 80% butyl acrylate and 20% 2-hydroxyethyl acrylate was synthesized via conventional free radical polymerization and diluted to 50% by weight in toluene. A more detailed description of the synthesis can be found elsewhere.¹³ A surface-modifying amphiphilic additive (SMAA) with a polysiloxane backbone of molecular weight between 900 and 1200 g/mol, and approximately 45–55% of the backbone consisting of chains of PEG-350, was synthesized according to previous work³⁷ and used in coatings formulations. Coatings were cast on both 76 × 152 mm² and 102 × 203 mm² aluminum panels (0.6 mm thick, type A, alloy 3003 H14) purchased from Q-lab. These panels were first sandblasted and primed with Intergard 264 marine primer, provided by AkzoNobel International Paint. Intersleek 700 (IS700), 900 (IS900), and 1100SR (IS1100SR) were used as commercial FR

coatings and provided by AkzoNobel International Paint. Dow Corning T2, a silicone elastomer, was used as a standard coating and was provided by Dow Corning. Falcon sterile polystyrene 24-multiwell plates were purchased from VWR International.

Experimental Approach. Previously, an amphiphilic siloxane-polyurethane (AMP-SiPU) FR coating had been reported by Galhenage et al.²³ This coating system had both poly(ethylene glycol) (PEG) and polydimethylsiloxane (PDMS) chains as the hydrophilic and hydrophobic components on a polyisocyanate pre-polymer backbone. Once the pre-polymers had been synthesized, they were mixed with an acrylic polyol and additional polyisocyanate and cured to form a self-stratifying SiPU FR coating. In that work, the amount of amphiphilic pre-polymer in the total formulation was varied, as well as the molecular weights of both the PDMS and PEG chains on the pre-polymer. The surfaces of the coatings were largely influenced by the molecular weights of both PDMS and PEG, with more significant heterogeneity resulting from higher molecular weights of PDMS. In addition, increasing the amount of amphiphilic modified pre-polymer in the formulation produced larger surface domains, also increasing surface heterogeneity. FR performance followed a similar trend, with higher removal occurring with higher molecular weights of PDMS and increased pre-polymer content in the formulation. Interestingly, similar trends were also observed for ice adhesion to the surfaces of these coatings.⁴⁴

To further investigate the surfaces and the FR properties of these coatings, two main objectives were established. The first was to vary the ratio of hydrophobic (PDMS) and hydrophilic (PEG) content in the pre-polymer composition and observe the changes in the surfaces of these coatings, along with their effect on FR properties. The initial AMP-SiPU chosen consisted of a 10 wt % PDMS and 10 wt % PEG isocyanate pre-polymer, resulting in a total of 20 wt % amphiphilic pre-polymer in the final formulation weight. The molecular weight of the PEG was 750 M_n , with the molecular weight of PDMS at 10,000 g/mol. This formulation was selected as it had one of the highest overall removal values for each marine fouling organism studied by Galhenage et al.²³ Changes in the PDMS/PEG ratio consisted of 5 wt % intervals about the base pre-polymer composition of 10 wt % of both PDMS and PEG. This resulted in formulations with a 20:0, 15:5, 10:10, 5:15, and 0:20 ratio of PDMS to PEG wt % in the pre-polymer.

The second objective of this study was to investigate the effects of incorporating a surface-modifying amphiphilic additive (SMAA) at a range of concentrations relative to the overall formulation of solids, into these AMP-SiPU FR coatings. It was expected that the amphiphilic nature of the SMAA would be more compatible with an amphiphilic coating matrix, potentially causing a variety of changes to surface morphology, as well as an increase in FR performance. The chosen SMAA for this study (Figure 1) consisted of a polysiloxane

partial comb-like architecture) caused significant changes in surface properties but did not produce surfaces with broad-spectrum AF/FR performance.³⁷ Thus, this type of SMAA was selected to determine if incorporation into an amphiphilic coating matrix could have synergistic effects in regards to the AF/FR properties of the system. The details on formulations prepared to investigate these methods of producing an amphiphilic coating system can be found in Table 1.

Table 1. Experimental Formulations Used in This Work Including the Ratios of PDMS to PEG (wt %) in the Pre-polymers, wt % of Added SMAA, and Total Hydrophobic and Hydrophilic Content from PDMS and PEG (wt %)

formulation	PDMS/PEG wt %	SMAA wt %	wt % PDMS (hydrophobic content)	wt % PEG (hydrophilic content)
F1	20:0	0	20.00	0.00
F2		5	21.25	3.75
F3		10	22.50	7.50
F4	15:5	0	15.00	5.00
F5		5	16.25	8.75
F6		10	17.50	12.50
F7	10:10	0	10.00	10.00
F8		5	11.25	13.75
F9		10	12.50	17.50
F10	5:15	0	5.00	15.00
F11		5	6.25	18.75
F12		10	7.50	22.50
F13	0:20	0	0.00	20.00
F14		5	1.25	23.75
F15		10	2.50	27.50

To determine the theoretical amphiphilic content of these formulations, the hydrophilic content of the additive had to be determined. This was accomplished by dividing the molecular weight (g/mol) of the grafted PEG chains by the total theoretical molecular weight (g/mol) of the SMAA, shown in eq 1

$$\% \text{ PEG} = \frac{\text{MW}\left(\frac{\text{g}}{\text{mol}}\right) \text{ grafted PEG chains}}{\text{total MW}\left(\frac{\text{g}}{\text{mol}}\right) \text{ of additive}} \quad (1)$$

The content was found to be 75% PEG and 25% PDMS. These values were used, in conjunction with the PDMS and PEG wt % in the pre-polymer compositions, to determine the theoretical hydrophobic and hydrophilic content based on total formulation solids of 15 g.

Amphiphilic Isocyanate Functional Pre-Polymer Synthesis.

For use in coatings formulations, an isophorone diisocyanate (IPDI) trimer (Desmodur Z4470 BA) was modified with monofunctional PDMS (10,000 g/mol) and hydroxy-terminated PEG (750 g/mol), resulting in amphiphilic pre-polymers. Several different pre-polymer compositions were synthesized as shown in Table 1. Detailed procedures for the synthesis of these pre-polymers can be found elsewhere.²³ However, a general procedure for the synthesis of the pre-polymer used in F9 (10 wt % PDMS and 10 wt % PEG based on total formulation solids) from Table 1 is as follows. To prepare 30 g of total formulation weight, PEG-750 (1.5000 g) was added to a 100 mL plastic cup after heating uncovered in an oven set at 100 °C for at least 1 h, along with ethyl-3-ethoxypropionate (EEP, 1.5000 g) and a Teflon coated magnetic stir bar. This mixture was stirred until PEG-750 was dissolved, and then PDMS (1.5000 g) was added and mixed vigorously with a vortex mixer for 3–5 min. Finally, IPDI trimer (1.6491 g) and catalyst solution (DBTDAc 1 wt % in MAK, 0.2079 g) were added to the mixture and again mixed vigorously with a vortex mixer for 3–5 min. Afterward, the plastic cup was sealed tightly with a lid, wrapped in parafilm, and the mixture was stirred overnight for 24 h. The isocyanate to total hydroxyl equivalents ratio was kept at 3:2

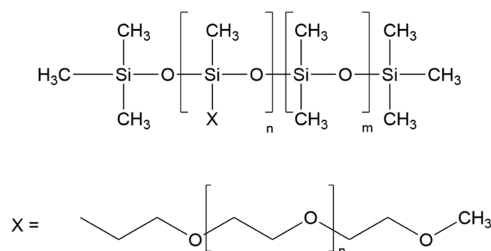


Figure 1. Structure of the SMAA used in AMP-SiPU FR coatings, where “X” is equivalent to a grafted PEG-350 chain on a polysiloxane backbone of ~900 to 1200 g/mol.

backbone of ~900 to 1200 g/mol, with ~45 to 55% of the backbone being grafted with chains of PEG with a molecular weight of 350 g/mol. This additive was one explored in previous work that investigated the changes in surface properties, and AF/FR properties of hydrophobic siloxane-polyurethane FR coatings resulting from SMAA incorporation. It was found that incorporation of this type of additive structure (with lower-molecular-weight PEG chains and

for all pre-polymer syntheses. A representative structure of these pre-polymers is shown in Figure 2.

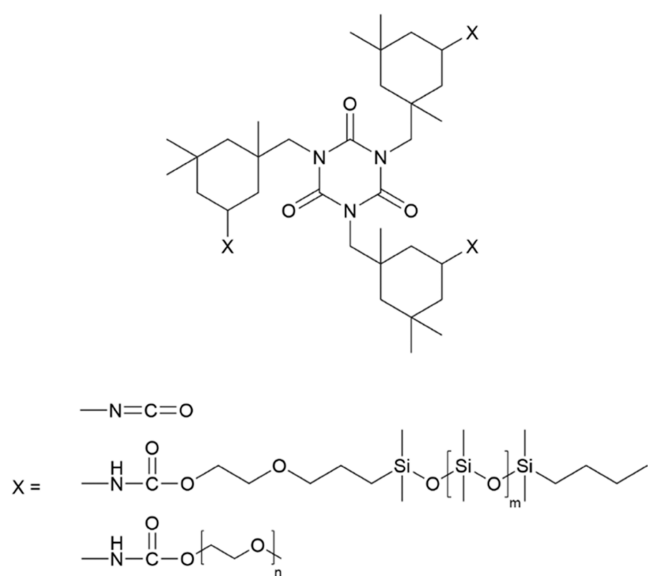


Figure 2. General structure of IPDI pre-polymers, where “X” can be isocyanate groups, chains of PDMS (10,000 g/mol), or chains of PEG (750 g/mol).

Pre-Polymer Isocyanate Titrations. To characterize the isocyanate groups after pre-polymer synthesis, isocyanate titrations were performed. These were performed after pre-polymer solutions had been stirred for 24 h and a general procedure, modified from ASTM D2572-19, is as follows. To a 250 mL Erlenmeyer flask, a 0.3–0.5 g sample of pre-polymer was added and diluted with 100 mL of isopropanol. Then, 25 mL of 0.1 N dibutylamine solution in toluene was added, followed by an additional 25 mL of isopropanol. This solution was mixed for 15 min, and then a few drops of bromophenol blue indicator solution was added. Using 0.1 N hydrochloric acid, the solution was titrated until an endpoint transition of blue to pale yellow. The amount of hydrochloric acid dispensed was recorded and a blank sample was also performed, omitting the addition of pre-polymer samples.²

$$\% \text{NCO} = \frac{4.202(B - S)N}{W \times \left(\frac{\% \text{ solids of sample}}{100} \right)} \quad (2)$$

Equation 2 was used to calculate the amount of remaining isocyanate content (% NCO) for the pre-polymer solutions, where *B* is equal to the volume (mL) of 0.1 N HCl used in a blank titration, *S* equal to the volume (mL) of 0.1 N HCl used in a sample titration, *W* is the sample weight (g), and *N* is the concentration of HCl (0.1).

Fourier Transform Infrared Spectroscopy. To further assess the presence of remaining isocyanate in the pre-polymer samples, FT-IR was performed using a Thermo Scientific Nicolet 8700 FT-IR spectrometer. Liquid pre-polymer samples were spread over a potassium bromide optical disk and placed in the instrument after obtaining a background spectrum.

DryAdd Simulations: Estimating Reactions between Primary Isocyanate and Hydroxyl Groups. To help determine the molecular distribution of the different pre-polymer structures, DryAdd Pro+, a Monte Carlo simulation package developed for polymer chain growth and network formation by Oxford Materials, was used. Generating predicted data from this type of software typically involves defining the materials present, defining the reactions that occur, and defining some general parameters of the reactions between the materials used. In this study, the materials used were a primary aliphatic polyisocyanate (IPDI trimer), carbinol-terminated PDMS, and hydroxy-terminated PEG. The IPDI trimer has three total

isocyanate groups, with PDMS and PEG containing one hydroxyl group each. The two reactions of interest were defined as the reaction between carbinol-terminated PDMS and IPDI trimer, and between the hydroxy-terminated PEG and IPDI trimer. The amounts, in grams and mols, of these materials were also entered into the software. Finally, the number of molecules simulated (population size) was set at 1 million, stopping at 100% conversion. Simulations of the formulations were performed and data about the distribution of different pre-polymer structures was obtained.

Gel Permeation Chromatography. A Tosoh Bioscience EcoSEC HLC-8320 GPC system was utilized to provide experimental evidence of a distribution of pre-polymer structures for data gathered from DryAdd simulations. To prepare samples for GPC analysis, leftover isocyanate was reacted with benzyl alcohol. Excess amounts of benzyl alcohol were added to the pre-polymers, solutions were heated to 60 °C, and stirred for an additional hour. Samples were then prepared by diluting with HPLC-grade THF and added to 1.5 mL sample vials. After thermal equilibrium, samples were passed through a TSKgel SuperH3000 column at a flow rate of 0.35 mL/min, with both column and pump temperature at 40 °C and pressure at 3.5 MPa. Chromatograms were recorded utilizing a refractive index detector.

Coating Formulation. Following pre-polymer synthesis, coating formulations were prepared as outlined by Galhenage et al.²³ A representative procedure for formulation F9 is as follows. After 24 h of stirring the pre-polymer solution, additional IPDI (5.5635 g), a 50% by wt in toluene, 80:20 BA/HEA acrylic polyol (14.4879 g), and pot-life extender acetylacetone (0.3000 g) was added to the mixture. The surface-modifying amphiphilic additive (1.6919 g), described in Figure 1, was added and the solution was mixed vigorously with a vortex mixer for 3–5 min, and then stirred, with the plastic cup sealed tightly, for an additional hour on a stir plate. The isocyanate to total hydroxyl equivalents was kept at 1.1:1.0. After 1 h, the formulations were cast on both 76 × 152 mm² and 102 × 203 mm² primed aluminum panels using a Gardco #80 wire wound drawdown bar, as well as on 75 × 25 mm² glass microscope slides. The coated panels and glass slides were left to cure at ambient temperatures for 24 h and then placed in an oven at 80 °C for 45 min. The remaining formulations were prepared in the same way, but with changes in wt % SMAA incorporation and composition of pre-polymer, as described in Table 1.

Control and Commercial Standard Coatings. As a comparison to the prepared experimental formulations, standard and control coatings were prepared for use in biological assays. These coatings (Table 2) were prepared according to the manufacturer’s specifica-

Table 2. Control and Commercial Standard Coatings Used in This Work

control name	control ID	composition
Polyurethane	PU	NDSU prepared polyurethane standard
Dow Corning T2	T2	silicone elastomer commercial FR standard coating
Polystyrene	Poly	polystyrene negative control
Intersleek 700	IS700	intersleek commercial FR standard coating
Intersleek 900	IS900	intersleek commercial FR standard coating
Intersleek 1100SR	IS1100SR	intersleek commercial FR standard coating with a “slime release” component

tions on 102 × 203 mm² aluminum panels primed with Intergard 264 and deposited in multiwell plates on primed aluminum disks.

Surface Characterization. A Thermo Scientific Nicolet 8700 FT-IR spectrometer, with a Smart iTR diamond plate ATR sampling accessory, was used to gather attenuated total reflectance Fourier transform infrared (ATR-FT-IR) spectra.

Water contact angle (WCA) and methylene iodide contact angle (MICA) measurements were gathered for all formulations using a First Ten Angstroms FTA100 series dynamic contact angle analyzer. Three measurements for each experimental formulation were taken at

2-min intervals for 10 min. A primed aluminum panel, coated with the experimental formulations, was placed on the sample platform and a 10 μ L drop of distilled water was carefully placed on the top, middle, and bottom portions of the panel. Measurements were started after a delay of 10 s. The values at each time interval were averaged and their standard deviation calculated. The Owens–Wendt method was utilized to calculate surface energy at each time interval over the course of 10 min and represented as mN/m.⁴⁵

Analysis of WCA, MICA, and surface energy calculations was also performed on experimental formulations after water aging. Coated primed aluminum panels were placed in circulating water tanks using tap water and left for 28 days at ambient temperatures. After drying the surfaces of coated samples, WCA, MICA, and surface energy values were determined using the same procedure described previously.

Atomic force microscopy (AFM) measurements were made following a general procedure. Sample panels were placed on the sample holder of an Oxford Asylum Jupiter XR microscope and controller. A silicon cantilever (AC240TS-R3) with resonant frequency of 50–90 kHz and a spring constant of 0.6–3.5 N/m from Oxford Instruments was affixed to the instrument and calibrated. An area of 100 $\times$ 100 μ m² was scanned in air and height and phase images were gathered. Images were analyzed using Igor Analysis software.

X-ray photoelectron spectroscopy (XPS) depth profiling measurements were performed following previously determined parameters. A Thermo Scientific K-Alpha X-ray photoelectron spectrometer, with a monochromatic Al K α (1486.68 eV) X-ray source and Ar⁺ ion source ($\leq$ 4000 eV), was utilized to determine the atomic concentration of C, O, and Si at the surface of the experimental coatings. An etch rate calibration step was also performed before depth profiling and a general procedure is as follows. A relatively thin film of polyurethane was prepared on a silicon wafer *via* spin coating at 6000 rpm for 35 s, using a Laurell WS-400A-6NPP spin coater. After 24 h at ambient conditions, the PU was fully cured in an oven set at 80 $^{\circ}$ C for 45 min and then a section of the coating was etched using a Trion Technology Phantom II Inductively Coupled Plasma etch system (ICP). After etching, the step height between the bare silicon surface and the PU coating was measured utilizing a KLA-Tencor profilometer to determine coating thickness. The coating thickness was determined to be 3.902 $\pm$ 0.111 μ m. To determine the etch rate through the PU, depth profiling was performed using a 1000 eV Ar⁺ source sputtering on an area of 400 $\times$ 400 μ m² in 1 min intervals for 30 min at a time until the silicon substrate was reached. Chamber pressure was maintained below 3.2 $\times$ 10^{−7} Torr, with photoemission lines for C 1s, N 1s, O 1s, and Si 2p collected after each etch for an interval of 10 s at a constant analyzer pass energy of 200 eV. The etch rate through the PU coating determined with this method was 0.1183 nm s^{−1}. After the etching calibration was performed, depth profiling experiments on free films of were performed using the parameters mentioned above. Samples were exposed to Ar⁺ sputtering for 30 etch cycles, with an average of three compositional scans represented for each cycle. Atomic concentrations were processed utilizing Thermo Advantage processing software.

Coatings Properties Testing. Crosshatch adhesion tests were performed according to ASTM 3359-17, using test method B. A cutting tool was used to make one patterned cut, with six teeth, and then making another cut going across the first pattern at 90 $^{\circ}$. The patterned area was inspected to ensure the cutting tool had cut through to the metal substrate, and then a strip of adhesive tape was flattened over the cut area, making sure air pockets had been removed, and then removed with constant force as close to 180 $^{\circ}$ as possible. The patterned area was then inspected for any loss of coating material and reported on a scale from 0 to 5B, with 0B indicating complete removal of coating material and 5B designating no removal of coating from the substrate.

Flexibility assessed *via* the conical mandrel bend method was performed according to ASTM D522-17. Samples were placed in the apparatus and bent to approx. 135 $^{\circ}$. Visual inspection of the bent samples was used to determine any visible cracking, and if cracking

occurred, the point at which the crack ended was reported in millimeters. This represents a measure of crack resistance and flexibility, with no cracking attributed to highly flexible coatings, with the increasing length of crack denoting reduced flexibility performance.

A modification of ASTM D5402-19 was used to determine the amount of double-rubs a coating could withstand until coating damage was observed. A hammer ($\sim$ 0.5 kg) was wrapped with 100% cotton cheesecloth around its head to form at least three layers and then soaked until dripping with methyl ethyl ketone. An area of 1" $\times$ 4" was selected on coated samples and the hammer was placed onto the surface and rubbed up and down (one double-rub) with minimal force until the substrate was seen. The cheesecloth was re-wet as needed, with new cheesecloth being replaced every 100 double-rubs.

The hardness of experimental coatings was also determined using pencil hardness according to ASTM D3363-20. In this method, samples were placed on a flat surface, and pencils with hardness values ranging from the hardest 8H to the softest 9B were pushed across the sample at a 45 $^{\circ}$ angle for a length of 0.25". The level of pencil hardness where there was no visible gouging of the coatings surface was recorded as the pencil hardness value. Coating thickness was determined using a BYK byko-test 8500 dry film thickness gauge. Three different locations (top, middle, bottom) of the coated samples were measured after calibration and reported as an average thickness.

Water Aging. Before submitting samples for biological assays, all formulations were placed in circulating water tanks for at least 28 days of pre-leaching at ambient temperature. This is an important step before performing biological assays. Residual toxic components of the coating system, such as dibutyltin diacetate, could potentially remain in the coating, resulting in errors during AF/FR evaluations due to leaching of these compounds. After pre-leaching for 28 days, panels were removed from water immersion, rinsed with deionized water, and dried overnight. Small disks were punched out from coated panels to be used in *C. lytica*, *N. incerta*, and *U. linza* biological assays. Larger, 102 $\times$ 203 mm² primed aluminum panels coated with formulations were used as is for *G. demissa* and *A. amphitrite* macrofouling assays.

Biological Assays. Several previously established biological assays were performed to evaluate the AF/FR performance of experimental coating systems compared to commercial standards and internal controls. These included growth and release assays involving the green macroalgae *U. linza*,^{46–48} cell attachment and release assays for the marine diatom *N. incerta*,^{46,48,49} *C. lytica* biofilm growth and adhesion assays,^{48–50} marine barnacle *A. Amphibalanus* reattachment and adhesion,^{51,52} and marine mussel *G. demissa* attachment and adhesion assays. Detailed procedures for these assays can be found in the cited references and brief summaries are provided in the [Supporting Information](#).

Further Analysis of *U. linza* Spore Settlement, and 7-Day Growth and Removal on Selected Formulations. Previous biological assays were carried out using the green algae *Ulva linza*, which measured sporeling growth and removal at water-jet pressures of 18, 67, and 110 kPa. To study the AF/FR properties of several selected formulations when subjected to much weaker hydrodynamic forces, a separate assay utilizing a turbulent flow cell was employed and a general description of the procedure is as follows.⁵³ Selected formulations included two different ratios of PDMS/PEG wt % in the pre-polymer, along with formulations containing SMAAs at 5 wt %. Coatings prepared on standard 75 $\times$ 25 mm² glass microscope slides were first equilibrated in 0.22 μ m filtered artificial seawater for 48 h prior to testing. To measure spore settlement (i.e., spore density), a suspension of zoospores (10 mL; 1 $\times$ 10⁶ spores mL^{−1}) was added to individual compartments of quadriPERM dishes containing the samples. Sample dishes were kept in the dark for 45 min at 20 $^{\circ}$ C, and then slides were washed by moving back and forth 10 times through a beaker of seawater to remove unsettled spores. Next, on one slide of each treatment group, attached spores were fixed using 2.5% glutaraldehyde in seawater. The density of zoospores attached to the surface was counted using an image analysis system attached to a fluorescence microscope. Spores were visualized by autofluorescence

of chlorophyll and counts were made for 30 fields of view (each 0.15 mm²) on each slide.

To measure sporeling growth, spores were settled on selected coatings via the same procedure used to measure spore density. After settlement, the spores were cultured using supplemented seawater medium for 7 days to produce sporelings (young plants) on 4 replicate slides for each selected coating sample. Sporeling growth medium was refreshed every 48 h. Sporeling biomass was determined *in situ* by measuring the fluorescence of chlorophyll contained within the sporelings in a Tecan fluorescence plate reader. The biomass was quantified in terms of relative fluorescence units (RFU). The RFU value for each slide is the mean of 70-point fluorescence readings taken from the central portion. Assessing sporeling removal was accomplished by placing these samples in a turbulent flow cell and subjecting them to 42 Pa of shear stress. Biomass remaining was determined using the same method used to determine sporeling growth. The percentage removal was calculated from readings taken before and after exposure to the shear stress.

Barnacle Cypris Larvae 48 h Settlement (*Balanus improvisus*). To perform settlement assays, cypris larvae of *Balanus improvisus* (= *Amphibalanus improvisus*) were obtained from broodstock that are maintained in a semicontinuous culture at Newcastle University, U.K. The original population of barnacles from which the broodstock has been cultured was from Sven Lovén Center for Marine Sciences, Tjärnö, Sweden. Adult barnacles were maintained in a recirculating aquarium kept at 19 °C, with brackish conditions, 25 ppt artificial seawater (ASW; Tropic Marin), and fed daily with *Artemia* sp. and *ad libitum* with the chlorophyte *Tetraselmis suecica*. Larvae were collected by removing adult barnacles overnight, and then immersing them, resulting in the release of nauplius larvae into the water column. These larvae were collected by attraction to a point light source. Nauplii (~10,000) were then transferred to 10 L buckets containing aerated ASW at a salinity of 25 ppt. Nauplii were fed with a mixture of 50:50 *T. suecica* and *Thalassiosira pseudonana* with the proportion of the latter being reduced to zero by the third day of culture. The metamorphosis into cypris took approximately 4–5 days when they were collected by filtration and stored in the dark at 6 °C for 3 days.

To analyze settlement of 3-day-old *B. improvisus* cyprids on selected coating samples, the general procedure is as follows.⁵³ Six replicate coated slides were placed in quadriPERM dishes, and 1 mL of ASW was initially added to each surface as a droplet. The cypris larvae were added 20 per drop, which had sufficient height for free movement of the larvae. The plates were then closed and covered in a moist tissue paper to minimize evaporation. Plates were incubated in the dark at 28 °C for 48 h. The number of larvae that settled on the samples was expressed as a proportion of the total, averaged per surface, and then compared between surfaces.

Statistical Analysis. One-way ANOVA was performed using Minitab statistical software on data gathered from the biological assays. These included growth and settlement assays for *U. linza*, *C. lytica*, and *N. incerta*, release assays for *C. lytica* and *N. incerta*, as well as reattachment assays involving *A. amphitrite*. The highest water/impact pressures, 138 kPa for *C. lytica* and *N. incerta*, were used for analysis. The results from these assays were treated as part of a completely randomized design, with two factors (pre-polymer composition and SMAA wt %) having five levels and three levels respectively for each factor, resulting in a total of fifteen treatment groups. The significance level (α) was set at $\alpha = 0.05$, with statistically significant results having *P*-values < 0.05. Tukey method was used to make comparisons between the difference of means from each treatment group. Means that do not share a letter in Tables S5–S22 are significantly different.

RESULTS AND DISCUSSION

The formulation of antifouling/fouling-release (AF/FR) coatings that offer protection from a wide variety of marine fouling organisms is a challenging prospect, largely due to the opposing surface preferences of these organisms. Recently, it has been shown that several coatings systems presenting an

amphiphilic surface consisting of both hydrophobic and hydrophilic domains perform better than traditional FR coatings systems. Two major methods of achieving this surface, as shown by Galhenage et al.²³ and Benda et al.,³⁷ involve formulating coatings with hydrophobic and hydrophilic domains within the coating's matrix itself, and also by incorporating noncovalently bound amphiphilic additives into a hydrophobic siloxane-polyurethane coating. Both methods reported an improvement in FR properties compared to coatings that did not have amphiphilic character and performed better than several commercial standard FR coatings. This study aimed to combine both methods of preparing an amphiphilic coating surface to investigate the effect on surface characteristics and AF/FR properties using biological assay testing with several well-known marine foulants. Several isophorone diisocyanate (IPDI)-based pre-polymers, modified with PDMS and PEG, were synthesized according to a previous study.²³ Five unique pre-polymers with different compositions were prepared by varying the relative amounts of PDMS and PEG, as shown in Table 1. Pre-polymers with PDMS and PEG molecular weights of 10,000 and 750 g/mol were chosen for this study because they showed superior FR performance compared to other compositions as reported by Galhenage et al.²³ These pre-polymers were combined with an additional IPDI trimer (Desmodur Z 4470 BA) and an acrylic polyol with an 80:20 ratio of butyl acrylate to hydroxyethyl acrylate to generate amphiphilic siloxane-polyurethane FR coatings (AMP-SiPU). In addition, a surface-modifying amphiphilic additive (SMAA), shown in Figure 1, was synthesized according to previous work and incorporated into these AMP-SiPU formulations at 5 and 10 wt %. The selected additive performed well, compared to other SiPU coatings with several other SMAAs with different structures, in biological FR assays reported by Benda et al.³⁷

To prepare isocyanate pre-polymers with varying PDMS and PEG compositions, an isophorone diisocyanate trimer (Desmodur Z 4470 BA) was reacted with 10k g/mol carbinol-terminated polydimethylsiloxane (PDMS) and 750 g/mol hydroxy-terminated poly(ethylene glycol) monomethyl ether (PEG) in the presence of dibutyltin diacetate solution in methyl amyl ketone. The ratio of isocyanate to total hydroxyl equivalents was kept constant at 3:2 to ensure leftover isocyanate necessary for subsequent coatings formulation. A modified isocyanate titration method based on ASTM D2572-19, along with FT-IR was used to determine the successful synthesis of pre-polymers.

Table 3 shows the theoretical and actual % solids and % NCO values of the pre-polymers with varying PDMS and PEG

Table 3. Theoretical and Actual Average Isocyanate (% NCO) and Nonvolatile Content (% solids) for Synthesized Pre-polymers with Varying PDMS and PEG wt % According to Table 1

formulation (PDMS/PEG)	theoretical		actual	
	% solids	% NCO	% solids $\pm$ SD	% NCO $\pm$ SD
F1 (20:0)	49.70	0.24	49.23 $\pm$ 0.18	0.09 $\pm$ 0.05
F4 (15:5)	51.38	0.74	49.75 $\pm$ 0.70	0.45 $\pm$ 0.05
F7 (10:10)	54.00	1.18	50.90 $\pm$ 0.27	0.83 $\pm$ 0.05
F10 (5:15)	54.80	1.50	52.11 $\pm$ 0.02	0.98 $\pm$ 0.05
F13 (0:20)	55.72	1.74	52.76 $\pm$ 0.18	1.15 $\pm$ 0.01

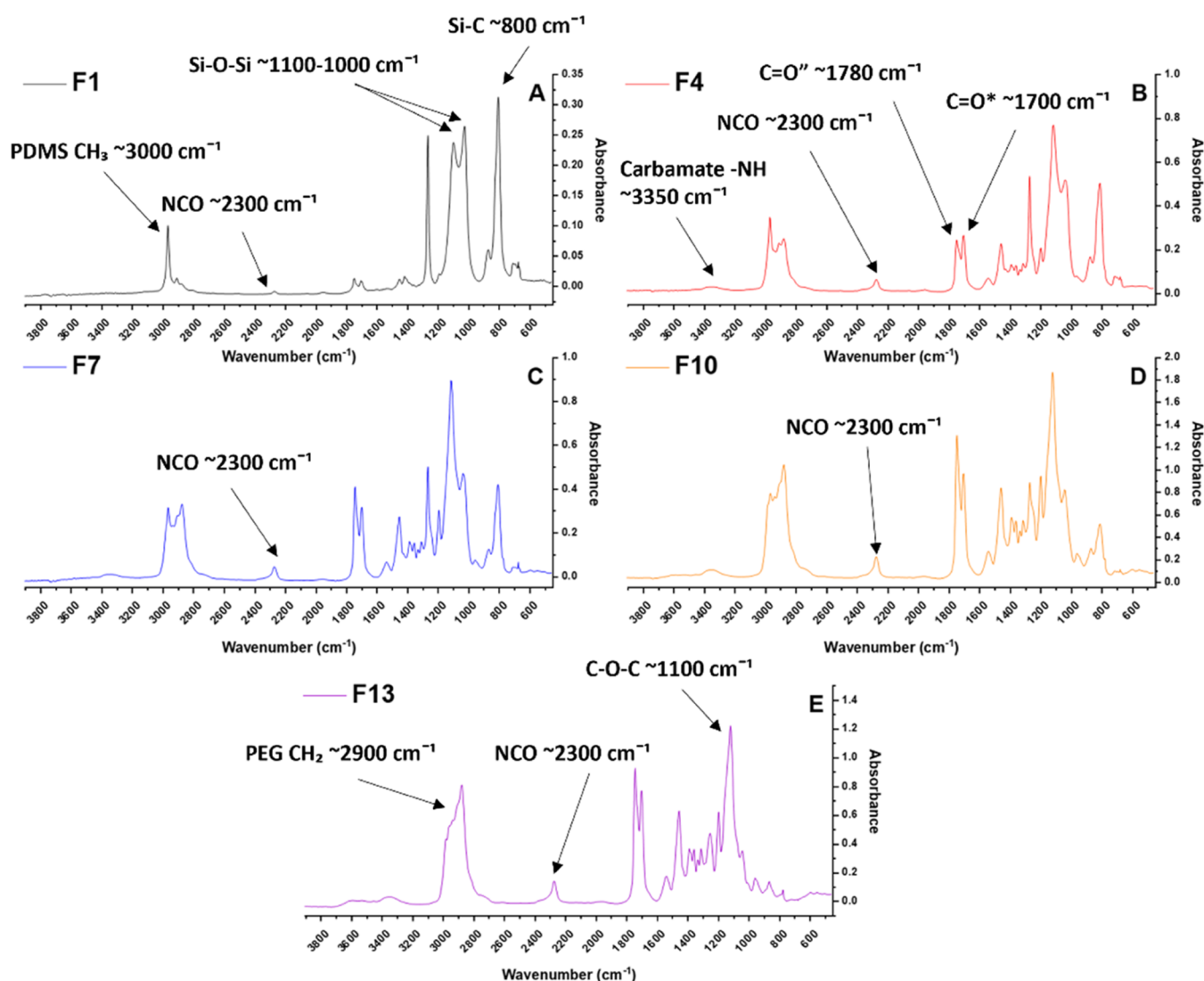


Figure 3. FTIR spectra A–E, pertaining to F1, F4, F7, F10, and F13, respectively, showing characteristic PDMS, PEG, carbamate linkage, and isocyanate peaks for pre-polymers shown in Table 3.

wt %. In each pre-polymer, a remaining amount of NCO sufficient for further crosslinking is evident. This suggests the successful synthesis of the desired pre-polymers. Deviations from the theoretical % NCO content could be explained by reaction of isocyanate with moisture, incomplete reaction with dibutylamine solution used during titrations, as well as a reduction in actual % solids, which will cause a decrease in the actual % NCO content.

Figure 3 shows the FT-IR spectra for the synthesized pre-polymers. All spectra (3A–E) show a peak around 2300 cm^{-1} , belonging to remaining isocyanate, necessary for further crosslinking reactions. In addition, peaks corresponding to the NH group of the carbamate linkage (reaction between isocyanate and the hydroxyl groups from the PDMS and PEG reactive species) can be seen between 3350 and 3450 cm^{-1} . The $\text{C}=\text{O}^*$ carbonyl peak appearing $\sim 1700\text{ cm}^{-1}$ corresponds to the carbonyl on the carbamate linkage, while the $\text{C}=\text{O}''$ carbonyl peak, near $\sim 1780\text{ cm}^{-1}$ results from the carbonyl on remaining isocyanate groups. In spectra 3A, belonging to F1 in Table 3 which only had 20 wt % PDMS in the pre-polymer, characteristic peaks around 3000 , 1100 – 1000 , and 800 cm^{-1} showed the presence of methyl groups

along the PDMS backbone, Si–O–Si stretching, and Si–C vibrations, respectively. Conversely, spectra 3E, belonging to F13 in Table 3, which only had 20 wt % PEG in the pre-polymer, showed peaks around 2900 and 1100 cm^{-1} belonging to methylene groups along the PEG chain as well as C–O–C ether stretching, respectively. Although the signature peaks for Si–O–Si and C–O–C overlap to some extent, the remaining spectra of 3B–D show gradual changes in composition based on the region of 1100 – 1000 cm^{-1} . Spectra 3B showed a more PDMS dominant region, while spectra 3D showed a more PEG dominant region. This is due to the change in PDMS/PEG ratio going from F4 at 15:5, to F10 at 5:15 shown in Table 3. Overall, more evidence of successful synthesis of these pre-polymers with varying PDMS and PEG wt % was obtained through FT-IR.

In addition to confirming the successful synthesis of amphiphilic isocyanate pre-polymers using FTIR and isocyanate titrations, we sought to understand the general structures in the pre-polymers. Due to the presence of three isocyanate functional groups on the IPDI trimer with the similar reactivity of the hydroxyl groups of the PDMS and PEG, a distribution of products is likely to form due to the

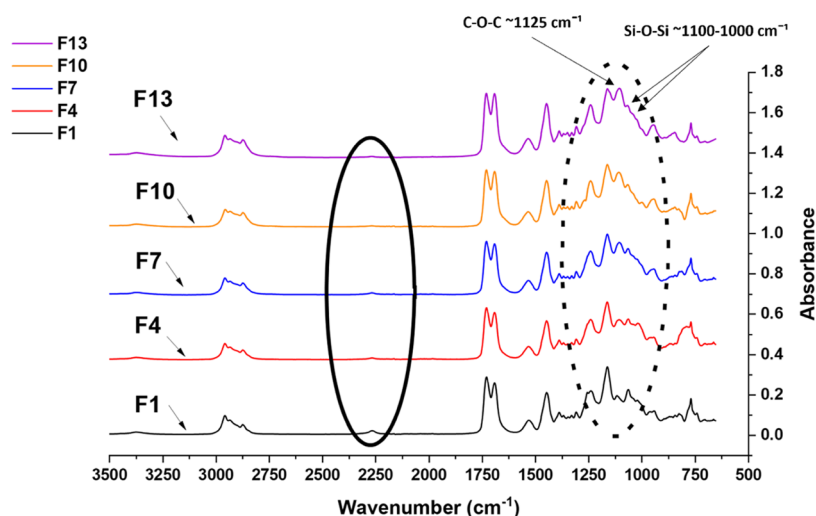


Figure 4. ATR-FTIR spectra for formulations shown in Table 1 with no incorporated SMAA. The solid black circle indicates the NCO region $\sim 2250\text{ cm}^{-1}$, while the dashed circle indicates the region containing Si–O–Si and C–O–C stretching ~ 1000 to 1125 cm^{-1} .

statistical nature of the isocyanate-hydroxyl reactions. While an “ideal” pre-polymer structure could be one that has one chain of PDMS, one chain of PEG, and one unreacted isocyanate group, there may be other structures such as an IPDI trimer fully reacted with PDMS, PEG, or combinations of both. To explore this hypothesis, we utilized DryAdd Pro+, a Monte Carlo simulation software package that can predict composition of molecular species during polymerizations or reactions between defined functional groups such as hydroxyl and isocyanate groups. After performing these simulations on the pre-polymers formed in Table 3, gel permeation chromatography (GPC) was carried out to determine if there was a correlation between computational and experimental findings.

Data gathered from DryAdd simulations, as well as GPC plots, can be found in Table S2 and Figure S1 in the Supporting Information. Overall, it was determined from DryAdd that for all of the pre-polymers, the number of reacted isocyanate groups of the IPDI trimer had a similar distribution, with $\sim 9\text{ mol } \%$ unreacted NCO trimer, $\sim 33\text{ mol } \%$ having one reacted NCO group, $\sim 41\text{ mol } \%$ having two reacted NCO groups, and finally $\sim 17\text{ mol } \%$ of a completely reacted NCO trimer. This range of pre-polymer species was not entirely unexpected as this was a simple reaction between an NCO and a primary alcohol. However, the data we gathered suggested that our generalized structure in Figure 2 is only slightly more dominant than the other species. What is more intriguing is the predicted abundance of fully reacted pre-polymer species, which would now be a noncovalently bound amphiphilic polymeric species that does not undergo reaction into the coating matrix. In Figure S1, GPC plots of the different pre-polymers with varying PDMS and PEG weight % show several peaks pertaining to pre-polymer species, which contain varying amounts of PDMS and PEG chains reacted onto the isocyanate trimer. It is clear to see that there is a shift in molecular weights for species containing an increased amount of PEG as opposed to PDMS chains on the pre-polymer. However, quantitative analysis was not performed using this technique and future work will be concerned with utilizing methods such as MALDI-TOF mass spectrometry to explore the fraction of these pre-polymer species to corroborate findings in DryAdd simulations.

After the pre-polymers were synthesized, coatings formulations were prepared as shown in Table 1. Surface characterization using ATR-FTIR, WCA/MICA, and surface energy calculations, as well as AFM were performed to investigate the effect of varying PDMS and PEG wt % in the pre-polymer as well as incorporation of varying levels of SMAA wt %. We hypothesized that as changes in the wt % of PDMS and PEG, and in the amount of SMAA incorporated in the coating, there would be a change in surface composition that could be detected via ATR-FTIR. Figure 4 shows the ATR-FTIR spectra for coatings F1, F4, F7, F10, and F13 described in Table 1. These coatings contained no SMAA and only had changes in PDMS and PEG wt % in the pre-polymer. The region for an isocyanate peak seen around 2250 cm^{-1} , shown by the solid black circle, is virtually absent in all formulations, signaling a fully crosslinked coating. The other region of interest, shown by the dashed circle, is between 1000 and 1125 cm^{-1} . This region contains an overlap between Si–O–Si stretching and C–O–C ether stretching indicative of PDMS and PEG, respectively. In F1, there was a calculated amount of PDMS at $20\text{ wt } \%$ in the pre-polymer, with $0\text{ wt } \%$ PEG. The characteristic “double” peak of Si–O–Si can be seen here, with no evidence of C–O–C from PEG. Contrary to this formulation, F13 contained $0\text{ wt } \%$ PDMS and $20\text{ wt } \%$ PEG. In this formulation, the surface shows the characteristic C–O–C ether stretching from PEG. For the rest of the formulations (F4, F7, and F10) various intensities of both Si–O–Si and C–O–C stretching peaks can be seen, with more PEG observed at the surface of F10 and more PDMS at the surface of F4.

Figure S2 (Supporting Information) shows the ATR-FTIR spectra obtained for formulations in Table 1 that had $10\text{ wt } \%$ SMAA incorporated into the coating (F3, F6, F9, F12, and F15). These spectra look like the ones shown in Figure 4, with the main difference being characteristic Si–O–Si and C–O–C peaks can be seen in all formulations regardless of pre-polymer composition. Overall, ATR-FTIR was able to observe the variation in PDMS and PEG content, from both changes in pre-polymer compositions and SMAA incorporation, at the surface of these coatings.

Measuring the water and methylene iodide contact angles (WCA/MICA) and then calculating surface energy (SE) can

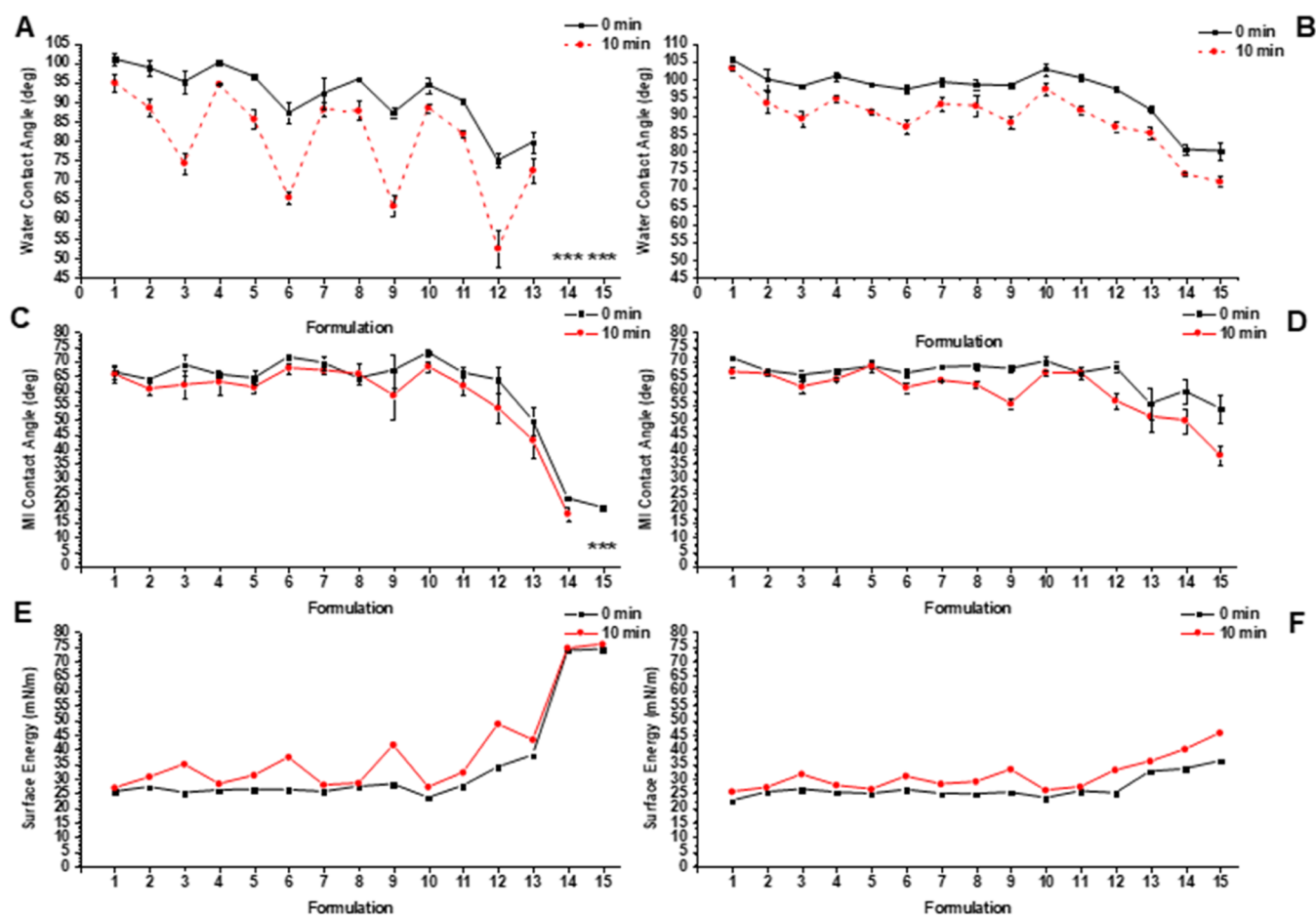


Figure 5. Water contact angles (WCA) (A, B), methylene iodide contact angles (MICA) (C, D), and surface energy calculations (SE) (E, F) on formulations shown in Table 1. A, C, and E were measurements performed before 28 days of water aging, with B, D, and F having been performed after water aging. *** in (A) and (C) denotes a completely wetted surface where no value could be obtained (effectively 0°).

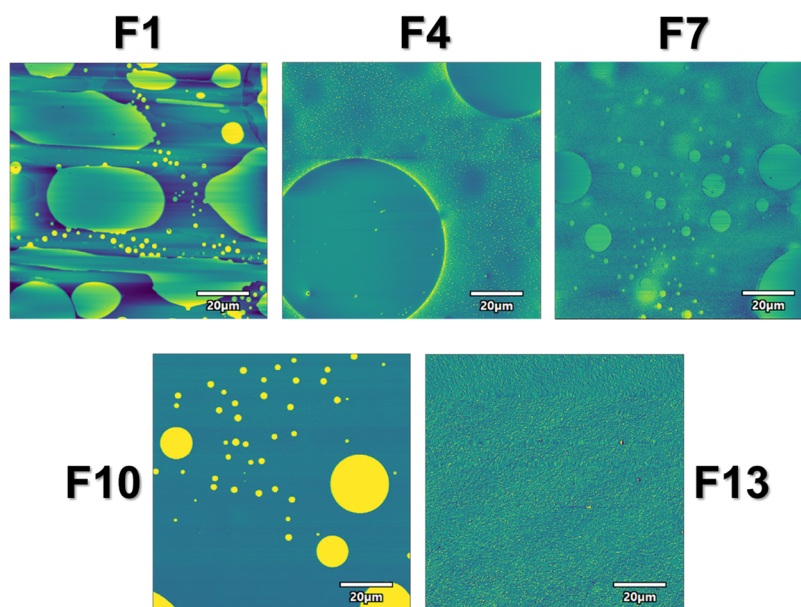


Figure 6. Phase images for AFM performed in air for formulations F1, F4, F7, F10, and F13. Each image is a $100 \times 100 \mu\text{m}^2$ scan of the coatings surface.

provide valuable information on the amphiphilic character of the surface, as well as surface activity due to SMAA incorporation for these formulations. Figure 5 shows the

WCA, MICA, and SE values for formulations in Table 1, before water aging (Figure 5A,C,E) and after water aging (Figure 5B,D,F). In each series, an increase in formulation

number going from 1 to 15, was associated with an overall increase in PEG content. The WCA values showed a slight downward trend when the amount of PEG in the pre-polymer composition was increased. These values all stayed above 90° , indicating that the surface of these coatings was still hydrophobic. However, as PDMS content was reduced in the pre-polymer composition, such as in F13, the WCA value dropped to $\sim 80^\circ$, becoming more hydrophilic. The same trend was seen in MICA and SE values, where pre-polymer composition had only a minor effect on surface character except for the fully hydrophilic pre-polymer compositions. Additionally, WCA, MICA, and SE values were not significantly changed (apart from the evaporation of the water droplet) after 10 min of analysis.

On the other hand, the incorporation of SMAA did have a significant effect on WCA values for coatings of all pre-polymer compositions. There was some decrease in WCA across formulations at the 0 min point of the measurements, with an insignificant effect on MICA and SE measurements. However, after 10 min of analysis, large decreases in WCA occurred for all formulations, with formulations containing 10 wt % SMAA having the most effect. Also, formulations F14 and F15, which had 5 and 10 wt % SMAA respectively, were considered completely wetted and no contact angle could be measured at either 0 or 10 min of analysis. This large decrease in contact angle over 10 min is not seen in MICA measurements. But SE calculations showed an increase in value after 10 min for formulations containing 5 and 10 wt % SMAA, with 10 wt % having the largest effect. These observations suggest that the presence of SMAA at the surface of these coatings has a more significant effect on surface hydrophilicity/hydrophobicity than changes in pre-polymer composition alone. The PEG chains grafted onto the additive modify the surface making it more hydrophilic under hydrated conditions, which causes a decrease in WCA and an increase in SE. The plots in SB, SD, and SF show the same WCA, MICA, and SE values, but they were performed after 28 days water aging. Overall, the changes in WCA and SE follow a similar trend to that recorded before immersion, but with a much-reduced effect. This is likely due to surface rearrangement of these coatings after immersion in circulating tap water. Studying the potential "leaching" of SMAA in formulations will be carried out in future work.

In addition to performing ATR-FTIR and WCA, MICA, and SE measurements, AFM was utilized to observe the surface morphology of formulations. Figure 6 shows AFM phase images of formulations F1, F4, F7, F10, and F13 which did not have any incorporated SMAA. Typically, in phase images, a surface with little difference in color would signal a largely homogeneous surface, with differences in color signaling a more heterogeneous surface. In the case of the surfaces presented in this work, more heterogeneous is defined as having both domains of PDMS and PEG on the surface of a polyurethane (PU) coating. A homogeneous coating can be seen as either consisting of PDMS on the PU surface, or just PEG alongside PU moieties. In F1, which has 20 wt % PDMS and no PEG in the pre-polymer, there are many irregular domains of PDMS that can be seen on the surface of the coating. Due to the low surface energy and hydrophobicity of PDMS, this was expected, as PDMS would prefer to not associate with a PU bulk. As the amount of PEG was increased throughout the formulations with differing pre-polymer composition, these domains begin to form more regular

shapes, showing a distinct difference in phase between PEG and PU moieties. Formulation F4 shows very large circular domains of PDMS with just a small increase in PEG content. In formulation F7, the concentration of PDMS and PEG are set to be equal at 10 wt % each. This results in surface morphology with smaller PDMS domains, spread more regularly throughout the coatings surface as there is not as significant a difference in amounts of the different materials present. Formulation F10 again shows quite distinct PDMS domains among a PEG and PU surface, but these are much smaller and less dense due to only containing 5 wt % PDMS in the pre-polymer. Finally, formulation F13 has a pre-polymer composition that has 20 wt % PEG and no PDMS, showing a phase image that is largely unchanged (homogeneous), consisting of PEG and PU moieties. From these phase images, increases and decreases in PDMS and PEG content in the pre-polymer composition have a significant effect on surface morphology.

In Figure 7, formulations F1 and F13 represent pre-polymer compositions of 20 wt % PDMS, 0 wt % PEG and 0 wt %

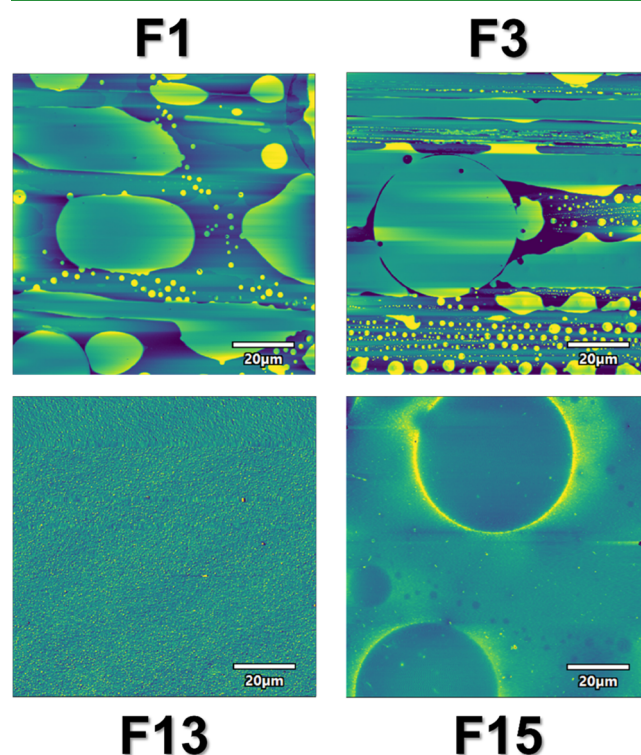


Figure 7. Phase images for AFM performed in air for formulations F1, F3, F13, and F15. F3 and F15 contain 10 wt % SMAA and have the same pre-polymer composition as F1 and F13, respectively. Each image is a $100 \times 100 \mu\text{m}^2$ scan of the coatings surface.

PDMS and 20 wt % PEG respectively, with no added SMAA. F3 and F15 represent those same pre-polymer compositions, but with 10 wt % incorporated SMAA. Comparing images of F1 and F3, as the SMAA was incorporated, the surface still showed large, irregular domains of PDMS on a PU surface. However, smaller, more regular domains are also seen, which are thought to be droplets of additive that have spread throughout the surface. This could be due to the addition of PEG via the SMAA making the surface even more heterogeneous. This effect is also seen between F13 and F15, where there is no PDMS in the pre-polymer of F13. But

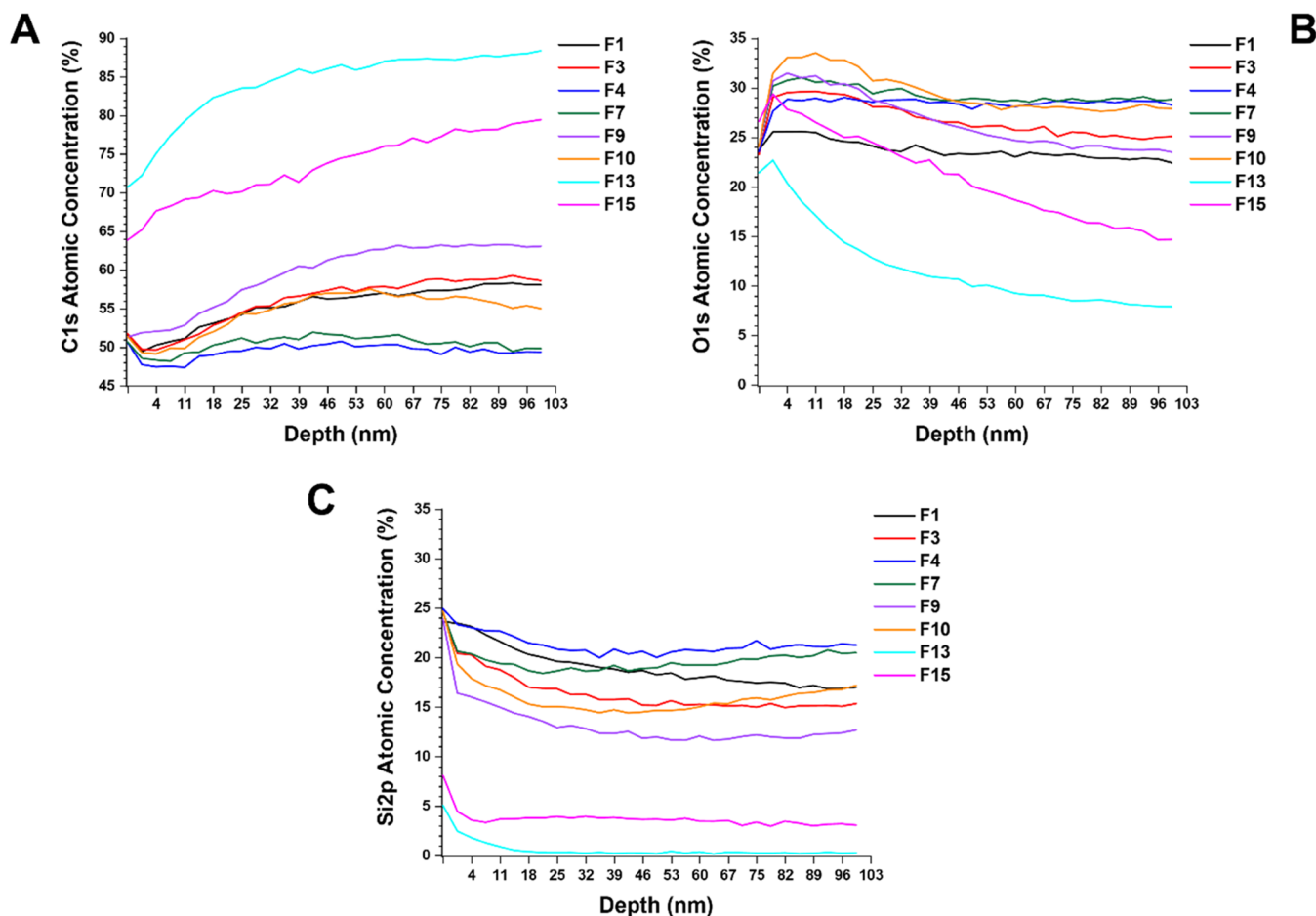


Figure 8. XPS depth profiling for selected formulations with different pre-polymer compositions and the presence of SMAA at 10 wt % concentrations, as shown in Table 1. Plots (A–C) correspond to the atomic concentrations (%) of C 1s, O 1s, and Si 2p atoms.

this time, the introduction of PDMS via the SMAA results in large, circular domains of PDMS aggregating on a largely PEG and PU surface. These phase images show that not only does the pre-polymer composition affect the surface morphology of these coatings, but that incorporation of SMAA does as well.

X-ray Photoelectron Spectroscopy (XPS) was utilized to determine the elemental composition of the experimental formulations when the samples were etched to a depth of ~100 nm into the surface of the coatings. The depth profiles obtained using this technique help to confirm the generation of a self-stratified, amphiphilic siloxane-polyurethane coating system. Figure 8A–C shows the atomic concentrations (%) of C 1s, O 1s, and Si 2p atoms plotted against etching depth (nm) for the tops of free films obtained after curing for selected experimental formulations with varying pre-polymer compositions as well as incorporation of SMAA at 10 wt % concentrations.

The formulations where the pre-polymer composition was varied to incorporate increasing amounts of PEG, and decreasing amounts of PDMS, are F1, F4, F7, F10, and F13 (Table 1). Here, it is noted that for pre-polymers containing appreciable amounts of PDMS wt % in the composition, there is a significant concentration of Si atoms at the surface, shown in plot 8C, which also extends at least 100 nm into the coating itself. This helps support the hypothesis that when PDMS is included in the pre-polymer, subsequent reaction with additional isocyanate and acrylic polyol results in a self-

stratified SiPU coating, with the PDMS being driven to the surface through differences in surface energies compared to other components such as the isocyanate and the acrylic polyol. There is little difference between atomic concentrations of pre-polymer formulations containing 20, 15, and 10 wt % of PDMS (F1, F4, and F7), with slightly lower amounts of Si at the surface with F10, a pre-polymer formulation containing only 5 wt % of PDMS in the composition. In addition, the Si atomic concentration for F13 is virtually nonexistent due to the absence of PDMS in the pre-polymer composition. Similar trends are seen in plots 8A and 8B, which detail the atomic concentrations of C 1s and O 1s. The oxygen concentration in the first 100 nm of the surface changes little between the formulations with increases in PEG in the pre-polymer concentration. However, there is still an increase in the concentration of O atoms compared to F1, which does not contain PEG in the pre-polymer, signaling the presence of PEG chains at the surface with PDMS. This provides further evidence that these coatings have a surface composition with an amphiphilic character.

Formulations incorporating SMAA were also analyzed (F3, F9, and F15). These coating formulations contained the additive at 10 wt % concentrations in pre-polymer formulations that had 20 wt % PDMS/0 wt % PEG, 10 wt % of both PDMS and PEG, and 0 wt % PDMS/20 wt % PEG in the pre-polymer compositions. For F3, the incorporation of additive significantly increases the amount of the atomic concentration of O,

due to the presence of PEG chains from the additive. In F9, the introduction of additive significantly decreased the concentration of Si at the surface, largely due to the increased amount of C and O from the PEG chains on the additive. Finally, in F15, the incorporation of additive significantly increased the Si atomic concentration due to the presence of PDMS from the backbone of the SMAA. Overall, it was shown that the surface compositions of these experimental coatings change significantly when differing amounts of PDMS and PEG are included in the pre-polymer composition. In addition, the incorporation of SMAA also affects the atomic concentrations of these amphiphilic SiPU coatings systems.

Several ASTM test methods, utilized to determine the physical properties of polymeric coatings, were performed on the experimental coatings. These tests were carried out to gain a better understanding of how the changes in pre-polymer composition, as well as incorporation of SMAA, affected properties such as adhesion to the substrate, chemical resistance, flexibility, and hardness. Results from these evaluations can be found in Table S22. The solvent resistance (toward methyl ethyl ketone) was negatively affected by the incorporation of SMAA into the coating matrix. This is largely due to the nonreactive nature of these additives, which when exposed to solvent, could be released from the coating matrix leading to the formation of voids at the surface. These voids could allow faster ingress of the solvent to the bulk/substrate interface, leading to rapid deterioration of protection. Also of note is the crosshatch adhesion for coatings which contained relatively high amounts of PDMS. Formulations F1 and F4 contain 20 and 15 wt % of PDMS in the pre-polymer respectively, and as SMAA is incorporated, there is a reduction in adhesion to the substrate. This could be due to the significantly higher amounts of PDMS present from both the pre-polymer and the SMAA. All experimental coatings exhibited high flexibility, with no cracking occurring during mandrel bend testing, largely in part to the flexible nature of PDMS chains, coupled with the chosen soft segment (acrylic polyol) used in these coatings. Additionally, coatings surfaces became softer as more SMAA was incorporated, again due in part to the nonreactive nature of these additives occupying spaces between crosslinks, which can lead to a sort of plasticizing effect on the coating matrix. Finally, measured thicknesses were mainly consistent across formulations, with the exception of coatings that contained only PEG or only PDMS in the pre-polymer. The reason behind the increased thicknesses for F13–F15 could be that as coating is deposited on the substrate, it was noted that there was a thicker buildup of formulation as the coating dried. This led to local increases of film build across the coating surface. Overall, it is important to evaluate the physical properties of these coating systems as they need to endure harsh marine environments to provide AF/FR protection. Care needs to be taken to balance the amount of hydrophobic (PDMS) and hydrophilic (PEG) content in these coatings to retain desirable properties.

Before laboratory assays were performed, panels coated with experimental formulations, as well as the control and commercial standard coatings in Table 2, were pre-leached in circulating water tanks for 28 days. After this time, leachate toxicity assessments for *C. lytica*, *N. incerta*, and *U. linza* were performed according to previously described methods.^{46,54} Figures S3–S5 detail the results from these assessments. Compared to a positive growth control, each organism tested showed nominal growth in overnight extracts (ASW and

nutrients) from the experimental coatings. This indicates that the experimental formulations were nontoxic after the pre-leaching step and that the results from the laboratory assays were due to surface effects and not compromised by toxic materials leaving the coatings.

Green macroalgae such as *Ulva linza* are recognized as widely distributed fouling organisms.⁵⁵ The settlement, growth, and adhesion of many marine fouling species, including *U. linza*, are significantly affected by the wetting and surface characteristics of the substrate.^{56,57} *U. linza* reproduces through the production of motile spores (ca. 5 μm in length), which travel through the seawater to locate a surface suitable for settlement. Once attached, the spores germinate and develop into sporelings (young plants). Spore settlement densities are typically lower on hydrophilic than hydrophobic surfaces but their attachment is stronger.⁵⁸ For surfaces with hydrophobic character, the opposite is true. Multiwell plates containing experimental formulations were exposed to spores of *U. linza* which grew into sporelings of approx. 100 μm in length before adhesion was evaluated.

Sporeling growth after 7 days, shown in Figure S6, did not show any significant trends with regard to pre-polymer composition. Coatings with pre-polymer compositions that contained both PDMS and PEG showed the lowest amounts of growth when SMAA was incorporated, with some performing as well as, or better than the commercial standard IS1100SR (Table S5). The adhesion of sporelings to the coatings was assessed by measuring the removal at three different water-jet impact pressures of 18, 67, and 110 kPa (Figure 9).

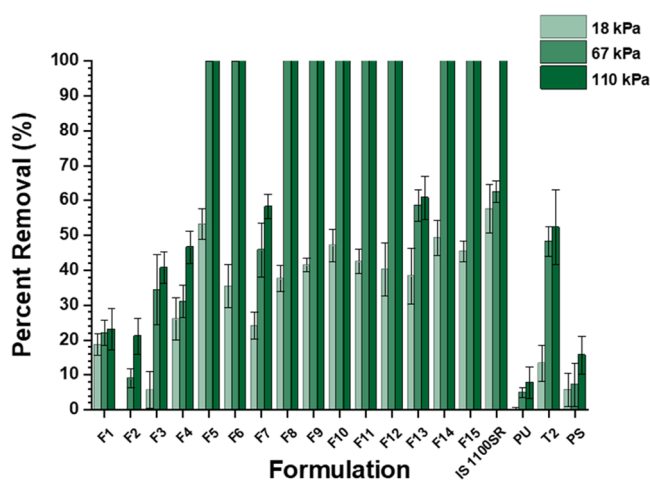


Figure 9. Removal of *U. linza* sporelings at water-jet treatments of 18, 67, and 110 kPa. Bars represent the average removal from six replicates with standard deviation. The X-axis is labeled to indicate the formulation number as described in Table 2, along with commercial standards and controls.

Formulations with varying PDMS and PEG concentrations in the pre-polymer all showed less than or equal to 60% removal at the highest water pressure. The formulations which contained more PEG in the pre-polymer had higher levels of removal than those with only PDMS, or low concentrations of PDMS, in the pre-polymer. This seems to deviate from the expected behavior of hydrophilic surfaces having lower removal. But, as seen through CA and AFM measurements, even the surfaces of coatings, which only contained PEG in the pre-polymer, showed a nonuniform surface with dynamic

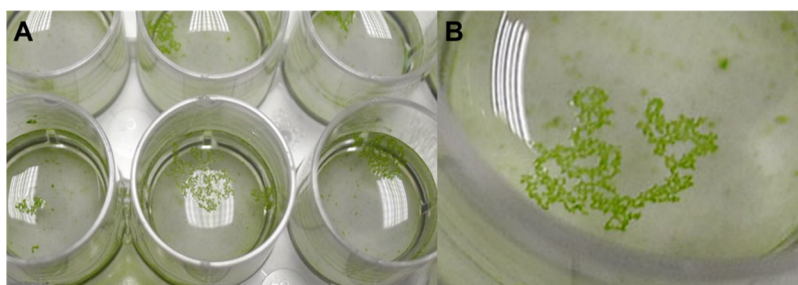


Figure 10. (A) Image of floating *U. linza* sporelings that have spontaneously released from coated surfaces in plastic 24-well plates. (B) Close-up of released sporelings.

surface activity. In this case, the addition of a hydrophilic moiety, such as PEG, to these coatings systems may improve release properties toward *U. linza*. Another significant and surprising observation was that for nearly every coating that contained the addition of SMAA to the formulations, even at 5 wt %, the sporelings spontaneously released from the surface upon disturbance of the dishes. The detached rafts of sporelings are shown in Figure 10.

The addition of SMAA also corresponds to an overall increase in PEG content within the coating, while also producing excellent removal in nearly every formulation. Again, this suggests that the notion of hydrophilic surfaces performing poorly in removal assays does not necessarily apply to these types of coatings. Overall, it was shown that changing the composition of the pre-polymer, by increasing the PEG content, improved removal performance, and that the inclusion of SMAAs had a significantly greater effect in increasing fouling-release performance toward the green alga *U. linza*. This result adds to the growing amount of evidence that shows improved AF/FR performance against this marine fouling species when amphiphilic moieties are utilized.^{59,60}

Experimental coatings were also subjected to the growth and adhesion of another marine alga, the slime-forming diatom *Navicula incerta*, and FR properties were evaluated. Typically, *N. incerta* prefers hydrophobic surfaces, especially those consisting of PDMS, to attach to and form a slime film.^{61,62} Previous work carried out by Benda et al. showed that as the surface of a PDMS-rich surface became more heterogeneous via the introduction of hydrophilic domains, cell attachment and adhesion decreased.³⁷ The types of surfaces explored in this work follow a similar trend. Figure 11 details the cell attachment and removal of *N. incerta* on experimental coatings, as well as controls and commercial standards. The highest cell attachment densities were on the **F1 coating**, which contained 20 wt % PDMS and 0 wt % PEG in the pre-polymer with no added SMAA. As the amount of PEG in the pre-polymer increased, only coating **F7** containing 10 wt % of both PDMS and PEG, showed significantly lower cell attachment comparable to the commercial standard **IS900** (Table S13). The most significant effect for coatings incorporating SMAA was seen for formulations **F1–F3**, containing 20 wt % PDMS in the pre-polymer. As more SMAA was added, significantly lower cell attachment occurred, with results comparable to the amphiphilic slime release commercial standard **IS1100SR** (Table S15). Overall, only a slight decrease in cell attachment was seen when PDMS and PEG contents were varied in the pre-polymer. However, more significant decreases in cell attachment were seen when the SMAA was incorporated, suggesting that a more heterogeneous, amphiphilic surface can

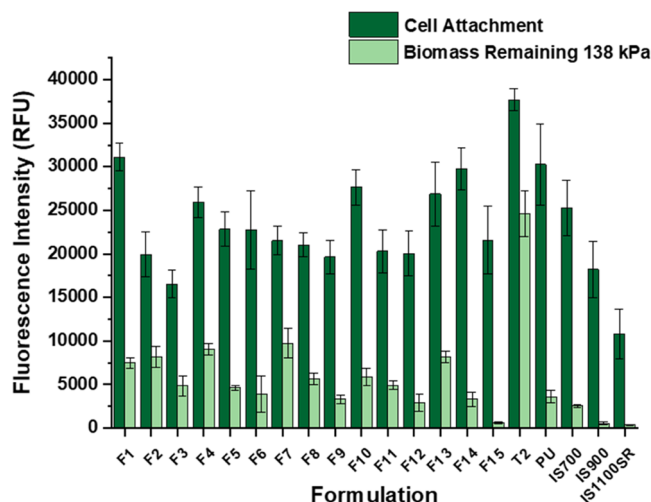


Figure 11. Cell attachment and biomass of *N. incerta* remaining after water-jet treatment at 138 kPa water column pressure. The dark green bars represent diatom cell attachment, while light green bars represent biomass remaining after water-jet treatment at 138 kPa for three replicate measurements and standard deviations. The X-axis is labeled to indicate the formulation number in Table 2, along with commercial standards and controls.

reduce diatom attachment, even in highly hydrophobic, PDMS-rich surfaces.

After water-jet treatment at 138 kPa, significantly lower biomass levels remained on all experimental formulations than on the **T2** silicone elastomer standard. However, there was no significant difference between removal with changes in PDMS and PEG content in the pre-polymer (Table S16). In contrast, the incorporation of SMAA resulted in a significant decrease in biomass remaining for each pre-polymer composition. It was shown that the inclusion of 10 wt % of SMAA resulted in the lowest amounts of biomass remaining, with higher removal performance occurring as the total PEG content in the coatings increased (Table S18). Coating formulation **F15**, with 20 wt % PEG in the pre-polymer, along with 10 wt % SMAA performed comparably to the commercial standard **IS1100SR**. Again, the incorporation of SMAA seemed to provide a more significant decrease in biomass remaining. This provides more evidence that the presence of this dynamic, amphiphilic surface layer helps to disrupt attachment and adhesion of *N. incerta*, facilitating its removal.^{63–65}

Another marine organism that can be a major contributor to marine biofouling, is the bacterium *Cellulophaga lytica*. Both hydrophobic and hydrophilic surfaces offer suitable surfaces for this marine bacterium to settle on, which can make it challenging to combat.⁷ Biofilm growth and retention assays

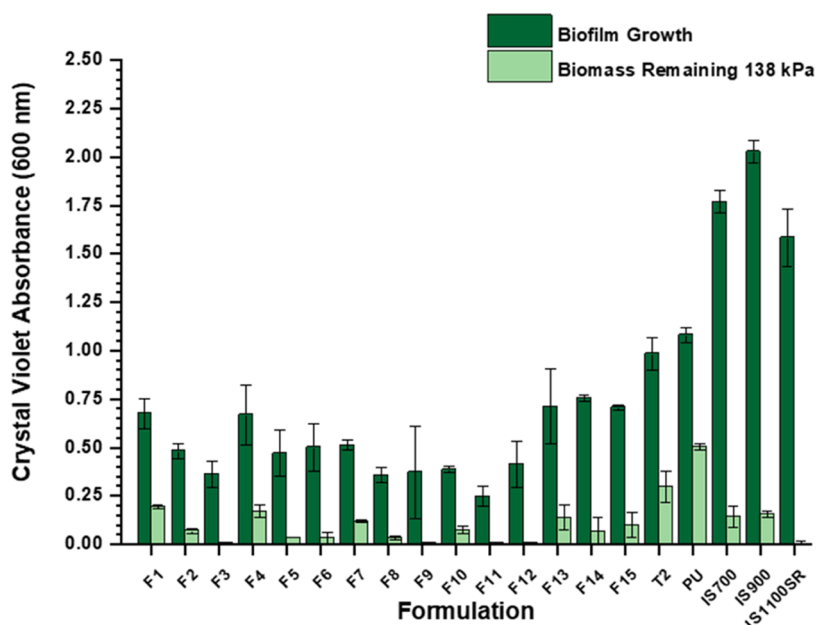


Figure 12. Bacterial biofilm growth and retention of *C. lytica* after water-jet treatment at 138 kPa water column pressure. Dark green bars represent biofilm growth, while light green bars represent biomass remaining after water-jet treatment at 138 kPa for three replicate measurements with standard deviations. The X-axis is labeled to indicate the formulation number in Table 2, along with commercial standards and controls.

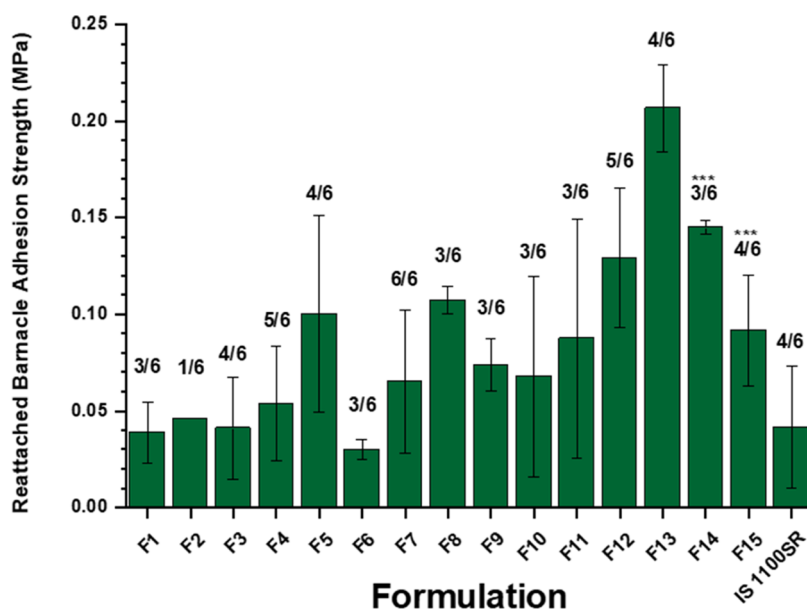


Figure 13. Reattached barnacle adhesion strength of *A. amphitrite*. Each bar represents the average adhesion strength of successfully pushed off barnacles. The ratio corresponds to the number of attached barnacles over the number of total available barnacles. *** denotes the breakage of an attached barnacle upon removal. The X-axis is labeled to indicate the formulation number in Table 2, along with commercial standards and controls.

were performed on experimental coatings, along with controls and commercial standards, and the results can be seen in Figure 12. Virtually all experimental coatings showed lower growth of the bacteria compared to all of the controls and commercial standards. Some formulations, especially those with both PDMS and PEG in the pre-polymers, performed significantly better (Table S8). Comparing the formulations in which the PDMS and PEG content varied in the pre-polymer, there was no significant difference in both growth and removal of bacterial biofilms (Tables S7 and S10). As SMAA was added to these formulations, only formulations F2 and F3 resulted in

significantly decreased biofilm growth (Table S9). However, removal of these biofilms was significantly greater when SMAA was incorporated into each formulation except for the purely PEG-containing pre-polymer formulations F13–F15 (Tables S12). Again, there is evidence that the coatings containing SMAAs had significantly lower growth and higher removal of *C. lytica*, despite its affinity for both hydrophobic and hydrophilic surfaces.

The biological assays with algae and bacteria were supplemented with macrofouling assays using the marine barnacle *Amphibalanus amphitrite* and marine mussel *Geukensia*

demissa. Fouling of ships' hulls with the macrofoulant *A. amphitrite* accounts for significant reductions in performance of marine vessels.^{9,66,67} Barnacles, in general, secrete specialized adhesives to adhere to surfaces. The adhesive of this particular species tends to spread more readily on hydrophilic surfaces, but still offers relatively high adhesion strengths on hydrophobic dominant surfaces.^{13,51,68} The force needed to push barnacles off surfaces is summarized in Figure 13 for all experimental coatings, including controls and commercial standards. Overall, there was a trend of increasing adhesion strength with increasing PEG in the pre-polymer (Tables S19). It is also seen that formulations F4, F7, and F10, containing a mixture of PDMS and PEG in their pre-polymers, did not show any significant difference in barnacle adhesion strength. When analyzing the formulations which had varying amounts of SMAA added to the formulations, again there was no significant difference between formulations with or without additive (Tables S21). The main finding of this assay was that formulations with high hydrophilic content tended to release the reattached barnacles less well than the more hydrophobic coatings. In some cases, the adhesive bond to the surface was so strong that the basal plates cracked. However, the performance of several formulations with more hydrophobic content was comparable to the commercial standard IS1100SR. The addition of SMAA did not significantly decrease FR performance and was shown to improve the performance of highly hydrophilic coatings surfaces, through the introduction of higher wt % PDMS in the formulation, as seen in F14 and F15.

The marine mussel *G. demissa* is another macrofouling organism that secretes several adhesive proteins called mussel foot-proteins. These mussels typically perform a "search" action to find a suitable substrate on which to deposit their adhesives, favoring hydrophilic surfaces over hydrophobic surfaces.⁶⁹ Figure S7 shows the mussel removal force, as well as overall number of attached byssal threads during pull-off adhesion measurements. Remarkably, no mussels attached to surfaces of formulations F1–F12, possibly due to sufficient heterogeneity of the surfaces caused by incorporation of SMAAs, as well as a mixture of PDMS and PEG in the pre-polymer. However, the mussels were able to attach to coatings that contained more hydrophilic content in the pre-polymer even with incorporated SMAA. It is likely that in these cases sufficient hydrophilic groups were presenting at the surface to make it favorable for the mussels to attach. Regardless of this, these coatings were shown to be highly effective in preventing mussel attachment and adhesion, which could prove useful in environments with higher populations of these marine foulants.

To gain a better understanding of those formulations that exhibited spontaneous release of *U. linza* sporelings, growth and removal assays utilizing a turbulent flow cell with significantly weaker shear forces was employed. Formulations F7, F8, F10, and F11 were selected to determine if there is an effect between different PDMS/PEG amounts in the pre-polymer (F7, F10), and how adding SMAA to the coatings may affect growth and removal (F8, F11). These coatings were cast on standard microscope slides (75 × 25 mm²) along with a glass control, and a PDMS elastomer standard (Bluesil). Results from these assays can be seen in Figure S8. The PDMS reference and glass control surfaces displayed very similar growth, being less than that of the experimental formulations. There was no significant difference in growth between F7 and F10, which contained differing amounts of PDMS and PEG in

the pre-polymer composition. However, the incorporation of SMAA in both pre-polymer compositions produced a significantly higher amount of growth on coated slides. In removal assays, at 42 Pa hydrodynamic shear force, there was virtually no removal from the glass control and PDMS standard surfaces. Formulations F7 and F10 show significantly greater amounts of biomass removal than the controls, with F7 performing better than F10. Again, when SMAA was incorporated (F8, F11), a significant increase in biomass removal was seen, with F8 showing removal of almost all sporelings that grew on the coated slide. Overall, despite formulations containing SMAA having considerably more sporeling growth, they still far outperformed all other formulations in the removal of sporelings at relatively weak hydrodynamic forces, making them attractive candidates for FR coatings.

Finally, settlement assays of *Balanus improvisus* cyprid larvae were carried out on the selected formulations (F7, F8, F10, F11), as well as glass and trimethoxy(octadecyl)silane coated substrates. As opposed to biological assays involving the reattachment of adult barnacles on the surfaces of experimental formulations, this cyprid settlement assay provides insight into how barnacles in earlier stages of their life cycle (cypris larvae) settle and ultimately undergo metamorphosis to adhere to the surface. The settlement of cypris larvae of barnacle species such as *B. improvisus* has been shown to be affected by surface chemistries, surface morphology, and overall surface wettability (hydrophilic vs. hydrophobic), with higher settlement typically seen on smooth, hydrophobic surfaces.^{68,70} Results from these settlement assays can be seen in Figure S9. The main observation was that virtually all of the available cyprids were unable to settle on these surfaces compared to ~95% settlement on a smooth hydrophobic surface, and ~25% settlement on a clean glass microscope slide (75 × 25 mm²). As larvae were performing their search actions for a suitable surface, an adsorptive effect was observed, where cyprids were immobilized but could not settle properly, and subsequently released from the surface. This effect was seen in all formulations but was most prominent in F8 and F11, which contained 5 wt % SMAA. It is thought that these surfaces were sufficiently heterogeneous in nature to deter larvae from attaching after contact with the initial "oil" layer brought about by the additives. Because of this effect, removal experiments could not be performed, but current efforts are underway to replicate this experiment and obtain useful removal values.

Overall, these biological assays provided valuable insight into the AF/FR performance of these amphiphilic siloxane-polyurethanes with varying PDMS and PEG content in the pre-polymer, as well as formulations containing surface-modifying amphiphilic additive. Growth and settlement of the marine foulants analyzed showed only slightly lower values than most of the commercial standards and controls, with only some of the formulations being comparable to IS1100SR. Growth tended to be lower on surfaces that had ratios of PDMS and PEG of 10:10 and 5:15 percent. However, inclusion of SMAA had a significantly larger effect on this growth than changing pre-polymer PDMS and PEG compositions in the experimental formulations. In each biological assay, formulations which contained SMAA, and had a mixture of PDMS and PEG in the pre-polymer composition, had lower growth values than other experimental formulations and control coatings, with some formulations performing comparable to IS1100SR. The FR properties of

these coatings share a similar trend. Changes in pre-polymer composition only showed a slight increase in performance, with formulations containing pre-polymer with a PDMS/PEG ratio of 10:10 and 5:15 again showing the best performance. When SMAA was incorporated, there was a significant increase in FR performance across all experimental formulations. Several formulations that contained SMAA at 10 wt % showed higher removal properties than even the commercial standard IS1100SR, with formulations having 5 wt % SMAA still performing comparably, if not slightly worse than commercial standards.

CONCLUSIONS

Synthesis of five different isocyanate pre-polymers with varying PDMS and PEG content was performed via established procedures. As the amount of PDMS decreased, and PEG increased, FTIR showed a decrease and increase in peaks associated with these moieties as well as the presence of isocyanate leftover from the reaction to form a crosslinked network. DryAdd simulations were performed to determine the distribution of pre-polymer species and showed that there was a wide range of pre-polymer species formed during synthesis. The highest fraction of pre-polymer species had two out of three available isocyanate groups reacted with either PDMS or PEG polymer chains. However, also present was a significant amount of completely unreacted isocyanate, as well as fully reacted isocyanate groups, which could possibly affect surface and AF/FR properties. GPC was performed to try and confirm these findings. Results from these experiments agreed with DryAdd simulations that there was indeed a broad range of pre-polymer species formed, but quantification was not possible and is currently being explored.

Formulations including these pre-polymers, with additional isocyanate, a synthesized acrylic polyol, acetylacetone, and catalyst solution were prepared and cast on primed aluminum panels. Additionally, a previously synthesized unreactive surface-modifying amphiphilic additive (SMAA) was incorporated into these pre-polymer formulations at 5 and 10 wt % solids. Coated samples were subjected to an array of surface characterization techniques to determine surface properties, as well as biological assays to assess AF/FR performance. ATR-FTIR was performed on all experimental formulations and showed an insignificant amount of isocyanate remaining, signaling cured coatings. In addition, formulations that did not contain SMAA again showed decreases and increases in peaks associated with PDMS and PEG moieties, displaying amphiphilic surfaces with varying hydrophobic and hydrophilic components. Peaks associated with PDMS and PEG were seen in all formulations which contained SMAAS, regardless of pre-polymer concentration, pointing to the SMAAs concentrating at the surface of these coatings. WCA/MICA measurements showed that there was only a slight drop in contact angles over 10 min as PEG content in the pre-polymer was increased. However, formulations that contained SMAA, especially at 10 wt % solids, showed significant drops in water contact angle after 10 min, suggesting that SMAAs present at the surface had a major effect on the wettability of the coatings surface. The highest decreases in WCA were seen in formulations with higher concentrations of PEG in the pre-polymer containing 10 wt % SMAA. AFM was also performed and collected phase images showed that formulations with only PDMS in the pre-polymer had a largely homogeneous surface morphology, with similar results seen in formulations with only PEG in the pre-

polymer. As these ratios were varied, large changes in surface morphology were observed, making the surfaces more heterogeneous via inclusion of both PDMS and PEG at the surface. Additionally, images were taken of surfaces that contained only PDMS or PEG in the pre-polymer, but also with 10 wt % incorporated SMAA. When SMAA was added, the surfaces also displayed large changes in morphology, again becoming more heterogeneous.

During XPS depth profiling measurements, further evidence of the formation of a self-stratified, amphiphilic siloxane-polyurethane (AMP-SiPU) was gathered. Changes in the PDMS and PEG concentrations in the pre-polymer composition, as well as incorporation of SMAA, significantly affected the C, O, and Si atomic concentrations up to 100 nm into the coating.

Several biological assays were performed to assess the AF/FR performance of these experimental formulations and compare them to controls and commercial standards. Settlement and growth of microfouling agents such as *C. lytica* and *N. incerta* were not significantly affected across all experimental formulations. However, FR properties were greatly enhanced compared to controls and commercial standards. Overall, it was shown that experimental formulations that contained SMAA, even at 5 wt % incorporation, exhibited a significant decrease in biomass remaining after water-jet procedures. Some experimental formulations, notably those with higher PEG content in the pre-polymer composition, while also incorporating SMAA, showed comparable, or better performance relative to commercial standards and controls in settlement and growth assays for the marine macroalgae *Ulva linza*. During removal assays for this alga, virtually every experimental formulation with SMAA displayed spontaneous release of settled sporelings upon shaking of test well plates. To investigate this effect further, several formulations were selected with and without SMAA incorporation, for use in growth and release assays of *U. linza* involving a turbulent flow cell with low hydrodynamic forces ~ 42 Pa. Although spontaneous release was not observed during this experiment, the incorporation of SMAA at only 5 wt % caused a significant increase in the removal of sporelings at low hydrodynamic forces. Barnacle reattachment and adhesion assays utilizing *Amphibalanus amphitrite* were also performed on experimental formulations and compared to commercial standard IS100SR. Overall, formulations with lower concentrations of PEG, either through pre-polymer composition or SMAA incorporation, showed lower amounts of barnacle reattachment, and lower removal strength comparable to IS1100SR. As PEG content was increased, stronger adhesion was observed. Additionally, cyprids of *Balanus improvisus* were used to evaluate the settlement and adhesion of selected formulations both with and without SMAA incorporation. These surfaces were shown to be excellent at resisting larval settlement, with each formulation having virtually no larvae settle on their surfaces, precluding removal experiments. Finally, marine mussel *Geukensia demissa* attachment and removal assays were performed and it was shown that no mussels could adhere to any of the experimental formulations except those with the highest amounts of PEG content. Overall, these coatings displayed broad-spectrum AF/FR performance in comparison to commercial standards and controls. The addition of SMAA had a more significant effect on both surface properties and FR properties than a change in PDMS and PEG wt % in the pre-polymer composition. Further work studying the interaction

between these additives and various other coatings systems is currently underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.2c10983>.

Details on the prepared pre-polymers and coating formulations; isocyanate titrations; DryAdd simulation data; GPC measurements for pre-polymer systems; detailed procedures for core biological assays; leachate toxicity assessments for *C. lytica*, *N. incerta*, and *U. linza*; spore settlement and sporeling growth for *U. linza*; settlement of *B. improvisus* cyprid larvae; additional growth and removal assays for on selected formulations for *U. linza*; and tables detailing one-way ANOVA results from biological assays (PDF)

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Notes

The authors declare no competing financial interest. The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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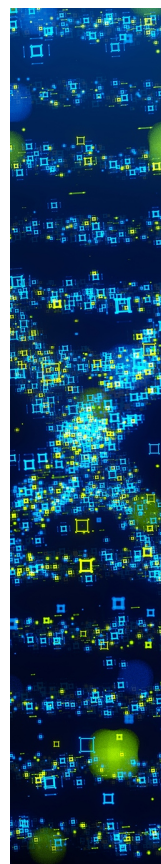
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